Game Developer’s Journal: Dahdi

Project: Dahdi – Traditional Indian Mills Board Game
Developer / Designer: Soumya Ranjan Samal
Development Period: 2026
Academic Project: College Game Development Project

Developer's Journal Manifesto

"A good game requires a sound strategy for success. The Game Developer's Journal serves as written evidence of strategic thinking, creative decisions, technical work, problem solving and refinement completed during development."

This journal records the major development stages represented by the final Dahdi project, including game selection, rule modeling, AI development, rendering, audio, accessibility and testing.

Journal Entry 1: Project Genesis & Game Selection

Phase: Concept & Research

The project began with the decision to create a traditional Indian-inspired board-game experience in a digital format. Dahdi was selected because the mills-style rule system has a clear board, a small number of pieces and meaningful strategic interaction. The goal was to keep the rules understandable while adding modern digital usability.

The initial design target was a browser game that could be demonstrated without a large installation. This led to a modular HTML/CSS/JavaScript architecture.

Journal Entry 2: Board Structure & Rule Modeling

Phase: Game Rules

The board was modeled as 24 playable points arranged across three concentric squares. The rules structure was separated from the visual interface so that placement, movement, mills, captures and win conditions could be tested independently. Nine pieces per player were defined as the standard setup.

Journal Entry 3: Placement, Movement & Flying

Phase: Core Gameplay

The game was divided into three phases. Placement allows players to put their pieces onto empty points. Movement allows legal adjacent movement after placement. Flying becomes available when a player has exactly three pieces, allowing movement to any empty point. This state-driven design makes the match flow predictable and easy to display in the UI.

Journal Entry 4: Mill Detection & Capturing

Phase: Rules Engine

The rules engine checks the mills passing through each board point. When a player forms a mill, the game identifies valid opponent pieces for removal. Protected pieces inside mills are handled according to the implemented capture rule, while a fallback is provided when all opponent pieces are protected.

Journal Entry 5: Artificial Intelligence

Phase: AI Development

A local AI opponent was added with three difficulty levels. Easy uses shallow search and some random choices. Medium uses depth-3 search and move ordering. Master uses deeper search with iterative/deeper analysis and a transposition cache. The evaluation function considers piece count, mills, threats and mobility.

Journal Entry 6: Rendering & User Feedback

Phase: Visual Development

The board renderer was separated from the rules engine. The interface highlights selectable pieces and legal destinations, shows the current phase and turn, and provides visual feedback when a match is won. Confetti was added as a lightweight celebratory effect.

Journal Entry 7: Audio System

Phase: Audio Development

The audio module was implemented using browser audio capabilities. It supports interface and gameplay sound effects and an optional Tanpura drone. Separate sound and music volumes allow the player to adjust the experience without changing the game rules.

Journal Entry 8: Settings, Localization & Accessibility

Phase: UX Development

The settings system was expanded with Player vs AI, Pass & Play, AI difficulty, diagonal board option, language selection, Tanpura music, SFX controls, colour-blind mode and reduced-motion mode. English, Hindi and Odia interface strings were included.

Journal Entry 9: Undo, Hint & Replay

Phase: Player Support

Undo allows the player to reverse an earlier action, while Hint uses the AI decision system to suggest a useful move. Rematch provides a quick route back into a new match after a result. These utilities improve experimentation and make the game easier to demonstrate.

Journal Entry 10: Testing & Final Refinement

Phase: Quality Assurance

Testing focused on placement, movement, flying, mill detection, protected captures, AI legality, UI controls, language settings, audio, accessibility and result states. The final refinement pass emphasized clear feedback and keeping the rules engine authoritative.

Current Project Reflection

Dahdi developed from a simple traditional board-game concept into a modular browser application with a rules engine, AI, rendering, audio and accessibility systems. The strongest technical lesson was the value of separating game rules from presentation. This separation made it easier to reason about legal moves, mill formation, captures and AI decisions.

The project also demonstrated that a small board can still create substantial strategic depth. The same 24 positions support placement tactics, blocking, mill threats, mobility decisions and a different endgame once Flying becomes available.

Final Development Note

The completed project provides a stable academic foundation. Future work can focus on guided tutorials, stronger visual animation, statistics, board themes and optional online multiplayer without replacing the existing core rules architecture.

Developer: Soumya Ranjan Samal
Project: Dahdi – Traditional Indian Mills Board Game
Year: 2026